

Master's Thesis — Agentic Pentesting (Hirschmann / Belden)

Full title: Master's thesis on the topic of Agentic Pentesting — Development and Evaluation of an AI Agent for Automating Penetration-Testing Workflows (m/f/d)

Company: Hirschmann Automation and Control GmbH (a Belden brand)

Department: Innovation · **Location:** Neckartenzlingen · **Type:** Full-time, fixed-term, Bachelor's / Master's / Diploma theses

Source: StepStone, requisition 14372911

Intro line from the posting:

"Together we are shaping the future of digitalisation and paving the way for the next generation of innovations. For the Innovation area of Hirschmann Automation and Control GmbH, location Neckartenzlingen, we are filling the following fixed-term position, starting immediately: Master's thesis on the topic of Agentic Pentesting — Development and Evaluation of an AI Agent for Automating Penetration-Testing Workflows (m/f/d)."

Your tasks

- Design and prototype an agentic workflow for pentest steps (tool orchestration)
- Integrate existing tools/scripts (e.g. Nmap, vulnerability scanner, script library)
- Use LLM/RAG for evaluating, prioritising, and plausibility-checking findings
- Build a simple human-in-the-loop interface for review/feedback and safe execution
- Define and carry out an evaluation against a baseline pentest (metrics + results)
- Whitebox code analysis and penetration-testing tools

Your profile

- Master's student in Computer Science, IT Security, or comparable
- **Good German and English skills**
- Solid Linux and networking fundamentals (TCP/IP) as well as security basics
- Programming experience (ideally Python)
- Interest/experience with LLMs, agentic workflows, and/or RAG
- Careful handling of sensitive data and security processes

Let's Create Possibilities Together (Belden boilerplate)

"Become part of a global community that is shaping the future of intelligent networking. At Belden you help push the boundaries of technology and write the next chapter of innovation — opening up new possibilities for your career, your future, and the world around you."

Notes. This is the substantive content captured from the StepStone listing; a live careers page may also carry benefits, "About us", data-protection, and equal-opportunity sections not captured here. **Language requirement:** the posting asks for "good German and English" — no CEFR level (no B2/C1) is stated.